

Cover letter

[SUBMISSION DATE]

Editors-in-Chief

International Journal of Climatology

Dear Editors,

Please consider our Research Article, “Distributional thermal change and the components of increasing dry–hot concurrence across Iran, 1991–2024,” for publication in the International Journal of Climatology.

The manuscript analyzes daily observations from 124 Iranian stations and addresses a question of broad climatological relevance: whether increasing dry–hot occurrence reflects changing marginal event frequencies, changing excess concurrence, or both. It combines fixed-network quantile trends in annual thermal-event counts with an exact station-level partition of observed dry–hot frequency change and synchronized block-resampling uncertainty.

Three findings define the contribution. First, thermal-count changes are consistently warming-oriented, while upper-minus-lower quantile and day–night asymmetry contrasts remain statistically unresolved. Second, annual and June–September dry–hot frequencies increase substantially on balanced station networks. Third, most of the summer point increase is associated with the hot-frequency term; the excess-joint term is close to zero and unresolved. The analysis also shows how zero precipitation thresholds and tie conventions alter interpretation in arid climates.

The work fits the journal's interest in observed climate-system change and in regional studies that inform broader climatological understanding. Its transferable contribution is a transparent framework for separating increased joint occurrence from altered excess concurrence while retaining network stability, threshold uncertainty, and structural-zero diagnostics. The manuscript deliberately avoids causal attribution and area-wide claims unsupported by the station design.

[AUTHORS TO CONFIRM BEFORE SUBMISSION: This manuscript is original, is not under consideration elsewhere, and has been approved by all authors. All authors satisfy the journal's authorship criteria. Any preprint is disclosed here: [NONE / DOI OR URL].]

[AUTHORS TO CONFIRM: The corresponding author has an ORCID and all required data, funding, conflict-of-interest, permission, and CRediT statements are complete.]

Thank you for considering this manuscript.

Sincerely,

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